Mobile-Based AI System for Lung Disease Diagnosis

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CERTIFICATE

This is to certify that this project report titled **"MOBILE-BASED AI SYSTEM FOR LUNG DISEASE DIAGNOSIS"** is a bonafide record of work done by **J. VEDA SAMEERA (228W1A1263), K CHANDI PRIYA (228W1A1227)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Masters of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26

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PROJECT SUMMARY

S.No	Item	Description
1	**Project Title**	**MOBILE-BASED AI SYSTEM FOR LUNG DISEASE DIAGNOSIS**
2	**Student Names & Numbers**	**J VEDA SAMEERA (228W1A1263)** **K CHANDI PRIYA (228W1A1227)**
3	**Name of The Guide**	**DR. Y. SANDEEP**
4	**Research Group**	**DEEP LEARNING**
5	**Application Area**	**HEALTHCARE**
6	**Aim of the Project**	The goal is to create a highly efficient, mobile-deployable deep learning system using MobileNetV2 capable of detecting and classifying lung diseases accurately from chest X-ray images. Besides accuracy, the model should also provide explainable visualization to help users understand the predictions and have the capability for automated report generation.
7	**Project Outcomes**	• Accurate detection and classification of lung diseases using deep learning models. • Development of a mobile-based system capable of generating predictions and visual explanations from chest X-ray images.

Student Signatures

1.
2.

Signature of the Guide

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LIST OF EQUATIONS

Equation	Brief description
Accuracy (3.2)	Accuracy that measures the overall correctness of the model
Precision (3.3)	Precision measures the proportion of true positive predictions among all positive predictions made by the model
Recall (3.4)	Recall measures the proportion of true positive predictions among all actual positive instances
F1 score (3.5)	The F1 score is the harmonic mean of precision and recall. It provides a balance between precision and recall
DenseNet Feature map (3.6)	Every layer takes the output of all preceding layers as input via concatenation.
EfficientNet Compound Scaling (3.7)	This formula uses constant scaling factors to scale network parameters equally
Grad-CAM Heatmap (3.8)	Computes a visualization by weighting feature maps with gradients
Softmax Function (3.9)	Acts as a normalization function that maps the input scores to probabilities

ABSTRACT

The importance of chest radiography in the early detection of lung diseases cannot be overstated, as it significantly supports the rapid diagnosis and treatment decision making process. Several diseases including bacterial and viral pneumonia, tuberculosis, and COVID-19 not only have to be accurately recognized but also at their initial stages in order to avoid their progression to severe stages. Generally, the diagnosis procedures depend on radiologists thoroughly and manually analyzing X-rays of the chest, which is not only an inefficient process but also prone to mistakes especially with overworked and expert-limited medical staff, a scenario quite common in rural healthcare settings. Artificial intelligence-driven computer-aided diagnostic systems can substantially contribute to the automation of lung disease classification based on chest X-ray images. Nevertheless, a majority of these systems still struggle with problems like decreased accuracy of their predictions, absence of sophisticated report generation, poor model explainability, and limited availability as desktop-only systems. Due to these factors, the current automated diagnostic solutions are having low levels of usability and trustworthiness. In order to mitigate these issues, the present study introduces a Mobile-Based AI System for Lung Disease Diagnosis, a project that embodies deep learning models of state-of-the-art along with modern techniques of optimization. To make the system transparent to a certain extent, explainability solutions like Grad-CAM have been incorporated, which also serve as aiding physicians by indicating possible disease locations in chest X-rays.

Keywords: Lung Disease Classification, Deep Learning, Model Optimization, Explainable AI, Generative AI Reporting, Chest X-ray Analysis, Mobile Health Application.

Chapter – 1

Introduction

1.1 Origin of the Problem

Diseases of the lungs like the bacterial and viral pneumonia, tuberculosis, and COVID-19 rank among top killers and causes of illness worldwide. Early diagnosis of these diseases can bring about timely treatments and lead to less serious complications of health. Among the diagnostic tools, chest X-rays hold a very important place since they are cost-effective, fast and are also widely available in hospitals and diagnostic centers.

On the other hand, diagnosing lung diseases in the traditional approach mainly involves radiologists manually interpreting chest X-ray images. This method is very slow, not only because it takes a lot of time but also one has to be very highly trained to identify disease patterns accurately. Experienced radiologists are in shortage, especially in many rural and less-developed areas, leading to delays in diagnosis and treatment. What's more, radiologists may sometimes make mistakes in the interpretation of medical images due to fatigue and heavy workload.

The advent of AI and Deep Learning technologies presents a great opportunity to help medical professionals in their work. These days, expert systems based on deep learning are helping medical professionals in analyzing images. For instance, CNNs are very good at not only detecting patterns from chest X-ray images but also at classifying various lung diseases. Nevertheless, many of the systems currently in use still

Bacterial pneumonia, viral pneumonia, tuberculosis, and COVID-19 are some of the lung diseases that are still the main causes of illness and death worldwide. These diseases torment millions of people globally each year, and the timely diagnosis is what will stop the occurrence of serious complications and avoid deaths. Among the diagnostic methods for detecting lung abnormalities, chest X-ray imaging is one of the most commonly used because it is cheap, fast, and available in most healthcare facilities. It enables medical professionals to

recognize the early-stage indications of infection, inflammation, and structural anomalies in the lungs.

Nonetheless, the conventional method for diagnosing lung diseases still remains largely dependent on the manual reading of chest X-ray images by exposed radiologists. Such a procedure necessitates considerable knowledge and hours to thoroughly examine each image and discover disease patterns. In fact, in many hospitals, especially those located in rural areas and developing countries, there is a shortage of skilled radiologists and medical staff. Consequently, the patients will have to endure the wait for diagnosis and treatment, and as a result, their health condition may worsen. On top of that, given the high level working load and time spent at work, radiologists suffering from fatigue that will undoubtedly increase human error and misinterpretation of X-ray images will be among those who can barely do their job.

1.2 Basic Definitions and Background

Modern computing technologies have evolved very fast, which has made it possible for healthcare sector to use artificial intelligence and deep learning techniques. Medical diagnosis systems have been significantly impacted by these technologies in terms of both accuracy and efficiency. Here, we provide some basic terms and background knowledge that support mobile-based AI system for lung disease diagnosis to be presented in this paper.

1.2.1 Artificial Intelligence (AI)

AI is a term which refers to developing software programs which have the capability to perform tasks requiring human intelligence. These tasks include being able to learn from examples, identifying regularities, making decisions, and working out difficult problems. AI has gained significant traction in the healthcare sector for automating medical image analysis, predicting diseases, supporting clinical decisions, and monitoring patients. These AI applications help medical practitioners by rapidly and accurately processing extensive medical data, thus leading to better diagnostics and treatment plans.

Systems based on AI technologies in medicine lessen the reliance on human labor and at the same time, reduce the risk of human errors. For example, when diagnosing lung diseases, AI tools are capable of interpreting chest X-rays and detecting visual differences linked to particular ailments. Thanks to these features, physicians are able to make their decisions more quickly and with greater certainty. Moreover, the integration of AI solutions in healthcare is also a step toward making medical services more reachable for the population living in rural or remote areas where expert radiologists may be scarce.

1.2.2 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) can be regarded as one of the deep learning structures dedicated specifically to the processing of imagery and tackling computer vision issues. A CNN is an assembly of numerous layers, including convolutional layers, pooling layers, and fully connected ones. These layers collectively enable the extraction of multi-level features from the input data. These are responsible for identifying basic elements such as edges, shapes, and textures, whereas pooling layers contribute by downsizing the feature maps and making the computations more efficient.

This work involves the utilization of cutting-edge CNN structures like DenseNet201 and EfficientNetB0 to categorize chest X-ray images visually into different lung disease classes. Thanks to their feature extraction capabilities, model's accuracy, and complexity management of medical image datasets, the above CNN models are more widely recognized and preferred for such tasks. Transitioning to CNN-based models leads to a considerable decrease in manual feature extraction and concurrently escalates the trustworthiness of automated disease detection systems driven by imaging.

1.2.3 Deep Learning

Deep Learning is a sophisticated field of artificial intelligence that mainly concentrates on the training of multi-layered neural networks. Through this training, these networks are capable of identifying very complex patterns in large datasets. Deep learning is extremely proficient when it comes to image, speech, and text data. The remarkable thing about deep learning

models is that they are capable of automatically determining significant features based on the raw input data. Hence, they do not need manually engineered features. This feature of deep learning makes it a great choice for carrying out medical image analysis like classification, detection, and segmentation.

Here, deep learning methods are utilized for the analysis of chest X-ray images and categorizing them into different types of diseases. By training on X-ray images, deep learning models are able to find detailed aspects like texture patterns and structural variations which are sometimes not easily visible to the human eye. This has the effect of increasing the accuracy of disease detection and minimizing the number of diagnostic mistakes.

1.2.4 Chest X-ray Imaging

Chest X-rays are a popular and effective technique to visualize lung health as well as heart and the chest cavity. These images are generated by passing x-ray radiation through the body and capturing the pattern of rays absorbed by organs, bones, and tissues. Besides being quick and inexpensive, chest radiography is the type of medical imaging most accessible worldwide. Radiologist examine these images to identify issues like infection, fluid in lungs, tumor, and scarring or other changes in lung tissues.

In this work, we focused on chest X-ray images to classify lung diseases. Our dataset contains X-rays labeled for different lung conditions like bacterial pneumonia, viral pneumonia tuberculosis COVID-19, and healthy lungs. Deep learning techniques are then applied on these images to uncover features of diseases and make correct classifications.

1.2.5 Explainable Artificial Intelligence (XAI)

Explainable Artificial Intelligence or XAI is a term used to describe methods that allow the inner workings of an AI model to be visible and understandable to users. It is a must-have for medical applications where the healthcare providers need to have a clear knowledge of how the decision-making of the AI system was done. By explaining AI, one can gain the trust of, be transparent with, and ensure the quality of the automated systems.

For instance, Grad-CAM (Gradient-weighted Class Activation Mapping) is one of the

most popular XAI techniques. It works by highlighting the important parts of an image that the model took account of for making its decision. Here, the project uses Grad-CAM to produce heatmaps that point out with visual aid the parts of the lungs in chest X-ray images that are diseased. Such a tool serves as a double-check by a user that the model is indeed zeroing in on correct lung areas which, in turn, raises a user's trust in the prediction results.

1.2.6 Generative Artificial Intelligence

Generative Artificial Intelligence refers to AI systems capable of generating original content in the form of text, images, or reports inspired by the input data. In the healthcare sector, generative AI might be employed to efficiently create medical reports by interpreting diagnostic results. Consequently, this not only lessens the burden on medical professionals but also maintains a certain level of uniformity in report creation.

For the present work, generative AI technology is able to write clinical style diagnostic reports from anticipated disease outcome and patient's data. These clinical reports give comprehensive details about the disease identified and suggest the ways to follow up symptoms/investigations. Furthermore, the use of generative AI increases the system's functionality by offering an automated documentation tool.

1.2.7 Mobile Health Systems (mHealth)

Mobile Health Systems or mHealth for short, is a term for the usage of mobile devices like smartphones or pods to provide healthcare and medical information. mHealth programs make it possible for people to get medical help even in very distant areas where there may be hardly any medical infrastructures. Mobile healthcare systems, in general, give users a possibility to upload medical info, get test results, and even interact with doctors.

Our AI system is set up as a mobile application through which users can upload chest X-rays and get a response with the prediction of whether they have a disease or not. Therefore, with this mobile solution diagnosis in real-time is possible and furthermore the system will be always accessible and at the users' disposal. These factors are definitely going to enhance both the functionality and the user-friendliness of the lung diseases diagnosis

system we have suggested.

1.2.8 Background of the Proposed System

The proposed system's background lies in the idea of combining deep learning models, explainable AI techniques, and m-health technologies to develop a holistic automated framework for lung disease diagnosis. This system merges state-of-the-art CNN networks, optimally trained models, and the features of generative AI, to deliver higher prediction accuracy and better user experience. The synergy of these technologies makes the proposed system a highly effective and proper solution capable of early detecting lung diseases.

1.3 Problem Statement

Lung diseases like pneumonia caused by bacteria or viruses, tuberculosis, and especially COVID-19 are still a huge threat to global health and the only way to do the right treatment is to diagnose these diseases in their early stages accurately. Chest X-ray is the most common tool for detecting changes in the lungs due to its low cost and high availability at healthcare facilities. Yet generally the diagnosis of lung diseases has been a slow and manual process where chest X-ray images are interpreted by radiologists only. Thinking of that manual process, it is really time-consuming and work-heavy, not to mention that it can lead to mistakes caused by tiredness, heavy work, and differences in how people interpret things.

That is not the case only in rural and remote areas but in many healthcare settings, there is a lack of trained radiologists and medical professionals. This deficiency results in a late diagnosis and treatment that can lead to worsening of the diseases and patient mortality. On top of that, most of the computer-aided diagnostic systems that are already in use do not have the requisite accuracy and reliability. It is because the models have not been optimized sufficiently and the training data are not adequate. There are systems which, apart from not being very accurate, do not explain in a clear way the reasoning behind their predictions; this running the risk of medical professionals being hesitant towards accepting automated decisions.

Yet another significant drawback of current diagnostic tools is that they do not come

with the functionality to generate clinical reports automatically. Usually, medical report writing is done manually which not only adds to the workload of healthcare professionals but also takes up a lot of their time.

1.3.1 Objectives

- The system delivers faster diagnoses by automating detection through deep learning on chest X-rays. This happens because algorithms trained on large datasets recognize patterns across multiple lung conditions, cutting down on time spent reviewing images manually.
- Model accuracy improves thanks to hyperparameter tuning and data addition, which reduce errors during prediction.
- Grad-CAM shows where abnormalities appear in the image, helping doctors see exactly what the model flagged, making decisions feel less abstract.
- Generative AI creates clinical reports automatically, saving hours every week in paperwork. The app runs on smartphones so patients can send scans and get feedback even in rural areas without clinics. Performance is tested using standard metrics like precision, recall, and confusion matrices to confirm reliability under real-world usage.

1.3.2 Outcomes

- A well-tuned, deep learning-based lung disease classification model that will be able to correctly identify various lung diseases such as bacterial pneumonia, viral pneumonia tuberculosis, COVID-19 along with normal cases.
- An interpretable decision support system in the diagnostic area which will generate visual heatmaps pinpointing the important regions in the chest X-ray images that lead to disease prediction.
- An un-supervised clinical documentation system which will produce well-organized diagnostic reports automatically on the basis of the model's predictions and patient details.
- A mobile platform that enables users to perform X-ray uploads, prediction viewing,

and real-time AI report receipt.

- Better access to diagnosis tools especially for remote and less-privileged areas, resulting in quicker medical decision-making and aiding healthcare workers.

1.4 Societal Applications of Proposed Work

Our suggested Mobile-Based AI System for Lung Disease Diagnosis covers wide-ranging possibilities that can directly impact society as a whole. Among other benefits, the system promises to enhance healthcare delivery, make medical diagnostics more widely available, and assist healthcare workers in their routine tasks. In essence, the understandable AI methods with mobile health devices has enabled the creation of the system that can effectively tackle healthcare problems faced by societies. Below are some of the most noteworthy ways our work can benefit the society at large.

- **Improved Healthcare in Rural and Remote Areas :**

The proposed system stands out as one of the most impactful societal applications capable of transforming healthcare services in rural and isolated areas. In fact, a severe lack of radiologists and medical experts to accurately decode chest X-ray images is a common occurrence in many underdeveloped regions. The residents of such areas very often have to cover great distances to get hold of suitable medical centers, thereby causing a delay in diagnosis and treatment. The AI diagnostic system based on mobile technology makes it possible for users to send chest X-ray images via a smartphone and get automated results in a very short time. This not only minimizes the reliance on specialists being physically present but also helps in providing diagnostic support more rapidly. Consequently, even in areas where healthcare facilities are hardly accessible, the early identification of lung diseases can be achieved**.**

- **Support for Telemedicine and Remote Diagnosis:**

In recent times, telemedicine has garnered a great deal of attention as a novel tool for providing healthcare services to individuals residing in distant locations. The system we propose for telemedicine supports the treatment and diagnosis of patients by providing

facilities for doctors and medical workers to analyze chest X-ray images without the patients having to physically attend the hospital. With the aid of the mobile platform, medical personnel will have the facility to examine the pictures that have been uploaded, scrutinize the interpretations of the AI system and suggest medical treatment to patients who are in different locations. This feature is highly advantageous during situations of widespread health scares such as epidemics when it is vital to reduce physical contact for the prevention of transmission. Embedding automated systems that detect diseases in telemedicine solutions raise the efficiency, accuracy, and speed of carrying out medical diagnoses to patients who are located remotely.

- **Early Detection and Prevention of Lung Diseases:**

The sooner lung diseases are detected, the better it is in terms of stopping the development of serious complications. A lot of lung problems like pneumonia, tuberculosis can be effectively cured if one discovers the symptoms very early. The proposed system offers automated interpretation of chest X-ray images and capable of identifying disease patterns very early, providing doctors with the opportunity to start the treatment immediately. When a patient is diagnosed early, there is a lesser chance that he/she will have to be hospitalized, the cost of treatment comes down, and the rate of recovery from the disease goes up. By encouraging early detection, the system helps in the improvement of the overall health of the community and at the same time, lessening the load on the healthcare institutions.

- **Reduction of Workload for Healthcare Professionals:**

Healthcare professionals, and radiologists in particular, are often subject to very heavy work schedules because of the huge influx of medical images that need to be analyzed. Manually interpreting chest X-rays is not only time-consuming but also demands sustained attention. Our AI-based system for radiology comes as a helping hand to professionals in the medical field by, first of all, automatically interpreting X-ray images and secondly by giving them their preliminary diagnostic results which they could always verify. This way, the time for carrying out the manual work is diminished and at the same time, the radiologists are given liberty to concentrate on those cases that are very detailed and require full evaluation. And

finally, the feature of automatically creating reports almost eliminates the time spent on the administrative work by generating structured clinical reports, thereby leading to enhanced efficiency within healthcare facilities.

- **Enhancement of Medical Education and Training:**

The system that is proposed has the potential to serve as a learning aid for medical students and healthcare trainees as well. The explainable AI component, that outlines the diseased areas in chest X-ray images, enables students to learn the visual representations of various lung diseases in radiographic images. Watching the heatmaps and classification outcomes produced by AI, learners can get a better understanding of the disease patterns and hone their diagnostic skills. This educational application is a great support for training programs and raises the quality of medical education. The improved training in the field of medicine with the help of AI technology helps the learners make better decisions by analyzing real-time data, and at the same time, it also offers simulations in the form of learning. The introduction of explainable AI also makes it easier for the learners to understand the patterns of diseases better.

- **Cost-Effective Healthcare Solutions:**

Global healthcare expenses are escalating globally, thus hindering many people from obtaining good medical treatment. The suggested AI diagnostic system based on mobile phones offers a cheaper way compared to usual diagnostic methods. The detection of diseases through automated processes minimizes the frequency of manual inspections and assists in making the best use of medical resources. Besides, early diagnosis can stop a disease from developing into the later stages, thus treatment costs are significantly reduced. Providing cheap diagnostic help is one of the ways through which the system can enhance the accessibility of healthcare services among the less fortunate people.

- **Improved Emergency and Pandemic Response:**

During emergency situations such as disease outbreaks or pandemics, the rapid diagnosis becomes a critical step in controlling the spread of infectious diseases. The system put forward could be utilized for prompt analysis of chest X-ray images and detection of respiratory

infections, which in turn, would facilitate the quicker isolation and treatment of affected persons. The mobile-based feature of the system allows healthcare teams to take it along even to the temporary medical camps and emergency response units. This feature of the system is what makes it capable of equipping the stakeholders to a state of readiness and assisting them in response to public health crises thereby leading to the better management of infectious diseases.

- **Promotion of Digital Healthcare Transformation:**

Employing mobile-based AI tools aids the digitization of healthcare services. Our system is an example of the blending of AI, cloud storage, and mobile apps for producing smart healthcare products. Such a move would pave the way for establishing advanced hospitals and e-health avenues that offer speedy, precise, and live medical assistance. Achieving great results with these technologies will pave the way for other methods, as well as updates in the health industry.

By implementing digital transformation in the field of healthcare using AI, we can have increased efficiency as the automation of various tasks will be easier and will ensure proper data handling. There will be greater connectivity between patients and healthcare practitioners via digital channels. The decision-making process will become easier due to data handling, which will result in more precise treatments.

Chapter-2
Review of Literature

This chapter contains a comprehensive analysis of the current research on automated identification of lung diseases through deep learning techniques on chest X-ray images. The goal of this literature survey is to grasp the recent techniques algorithms datasets, and issues in diagnosing lung diseases using artificial intelligence. The review covers papers from journals, conference proceedings, and medical imaging periodicals that are centered on the aspects of disease detection accuracy, interpretability, and the development of diagnostic systems that can be operated in real-time.

Many research papers examined CNNs and transfer learning techniques for the detection of pneumonia tuberculosis COVID-19, and lung cancer. Initial diagnosis methods depended upon manual analysis of chest X-ray images or traditional image processing techniques. Although these methods to some extent offered help, they were time-consuming, prone to errors, and demanded highly skilled radiologists. In many parts of the world, particularly those that are rural and have limited resources, the shortage of expert radiolo gists has led to delays in diagnosis and decreased the potential for effective treatment.

As deep learning technologies hit the scene, researchers started to build computer systems that could automatically identify complicated patterns in huge collections of medical data. These systems even outperformed human experts. Publicly available large-scale medical image datasets like ChestX-ray8 encouraged researchers to develop deep learning models, which resulted in greater dependability and better results of these models. Yet, some issues such as variations in the dataset, and the introduction of the model into real life still remain.

The works discussed in this chapter reveal different lung disease classification systems, including MobileNet, EfficientNet, and CNN hybrid architectures, that have made their mark in this field. Also, explainable AI methods, like Grad-CAM and SHAP, are extensively employed for the visualization of the disease areas and for the enhancement of the trust in AI predictions.

2.1 Description of Existing Systems

Publication details	Datasets details	Algorithms used	Purpose	Remarks
Deep Learning for Lung Disease Classification Using Transfer Learning and a Customized CNN Architecture with Attention **Authors**: H.Wedisinghe and T.G.I.Fernando **Type**:ICARC - Conference paper(IEEE)	Lung X-ray Image Dataset containing images categorized into healthy lungs, lung opacities, and viral pneumonia.	Transfer learning with SqueezeNet, VGG11, ResNet18, DenseNet, and MobileNetV2	Classify lung diseases using transfer learning and attention-based CNN.	Attention mechanisms improved feature localization and interpretability of the model.
AI-Assisted Severity Assessment of COVID-19 from Chest X-rays Using Explainable Deep Learning Models **Authors:**T.Ruga,E.Vocaturo,E.Zumpano **Type**:BIBM-conference paper(IEEE)	COVID-19 Chest X-ray Dataset (publicly available)	Deep Learning Models (e.g., DenseNet121), Explainable AI techniques (e.g., Grad-CAM), Severity Scoring Algorithms	Assess COVID-19 severity from chest X-rays using explainable AI.	Explainable predictions improved model transparency and helped integrate AI into clinical workflows.
Lung Cancer Detection Using Deep Learning and Explainable Artificial Intelligence Models **Authors**:A.Alomar,M.Alazzam,H.Mustafa,A.Mustafa **Type**:ICICS-Conference paper(IEEE)	LIDC-IDRI dataset (Lung Image Database Consortium and Image Database Resource Initiative)	Deep Learning Models (InceptionV3, ResNet50), Explainable AI techniques (e.g., LIME, Grad-CAM) .	To detect lung cancer accurately using deep learning models enhanced with explainable AI.	Corporates Grad-CAM for interpretability, aiding clinicians in understanding model decisions and improving trust in predictions.

Publication details	Datasets details	Algorithms used	Purpose	Remarks
Unified Synergistic Deep Learning Framework for Multimodal 2-D and 3-D Imaging Data in Lung Infection Detection **Authors:**M.Owais, M.Zubair, L.Seneviratne,N.Werghi, I.Hussain **Type:**Journal paper(IEEE Journal)	Multimodal imaging datasets combining 2- D (X-ray) and 3-D (CT) images	Synergistic Deep Learning Framework integrating 2-D and 3-D CNNs, Attention Mechanisms	To develop a hybrid framework combining 2D (X-rays) and 3D (CT) imaging for more effective lung infection diagnosis.	Enhances diagnostic accuracy by leveraging complementary features from both 2D and 3D modalities, useful in COVID-19 detection.
Explainable Deep Learning for Chest X-Ray Classification. **Author:**Akter S.et al **Type**:Journal paper (Non-IEEE Journal,MDPI)	NIH ChestX-ray8 Dataset (100,000 images)	Convolutional Neural Networks (CNNs), Transfer Learning, Grad-CAM	To classify chest diseases using deep learning models with explainable visual outputs for transparency.	Promotes interpretability using heat maps like Grad-CAM, making deep models more suitable for clinical applications.
Lung Disease Classification using X-Ray Imaging with Ensemble Learning. **Authors**:K.Vij,S.Thakur,A.Sharma,R.Mohana **Type:**ISED-Conference Paper(IEEE)	The study utilized the NIH Clinical Center's ChestX-ray8 dataset, which includes over 100,000 chest X-ray images.	Convolutional neural networks (CNNs) for feature extraction and support vector machines (SVMs) for classification	To improve lung disease classification performance using ensemble models combining multiple CNNs.	Ensemble learning improves robustness and overall accuracy by integrating diverse model predictions for better reliability.

Explainable AI for Early Lung Cancer Detection: A Path to Confidence **Authors**:X.Liu,Z.yu,L.Tan **Type**:ICSECE-Conference Paper(IEEE)	Lung CT imaging dataset, consisting of chest CT scan images of both lung cancer–positive and normal cases	NIH ChestX-ray8 Dataset (100,000 images)	Convolutional Neural Networks (CNNs), Transfer Learning, Grad-CAM for explainability	Achieved improved detection accuracy for early-stage lung cancer. Grad-CAM provided visual explanations.

Table 2.1: Description of Existing systems

Table 2.1 shows the Description of existing systems from different articles and from different sources.

2.2 Summary of Literature Study

- The research [1] on lung disease classification using transfer learning and customized CNN architectures highlights the effectiveness of models like MobileNetV2, achieving the highest accuracy (88.5%) by utilizing attention layers and pre-trained networks.
- The study [2] on AI-assisted COVID-19 assessment from chest X-rays applies explainable deep learning models (e.g., DenseNet121, Grad-CAM) to provide severity scores and visual explanations, aiding clinical decision-making.
- The work [3] on lung cancer detection combines deep learning models (InceptionV3, ResNet50) with explainable AI tools (LIME, Grad-CAM), achieving high sensitivity and specificity while enhancing clinical validation.
- The paper [4] presents a synergistic deep learning framework integrating 2D and 3D imaging data with attention mechanisms, significantly improving lung infection detection accuracy.
- The study [5] on explainable deep learning for chest X-rays uses SHAP analysis with models like ResNet50 to identify important features influencing predictions and improve disease classification.
- The research [6] on ensemble learning for lung disease classification shows that combining CNNs with SVMs outperforms individual models, highlighting the strength of ensemble approaches in medical imaging.
- The work [7] on early-stage lung cancer detection emphasizes transfer learning and Grad-CAM for better accuracy and interpretability, helping clinicians trust AI-driven diagnoses.

The literature review done for this project gave us a lot of useful information about the pros and cons of the in-use lung disease diagnosis systems. From the research articles that have been discussed, a few main points can be highlighted.

The very first research revealed that transfer learning and CNN architectures tailored to the lung disease scenario are a very good means of classification. For instance, MobileNetV2 has figured out how to get better accuracy levels by using networks that were trained beforehand and also attention layers. It has been shown through this research that classification performance can be significantly increased by the feature extraction method.

A different research targeted AI-assisted COVID-19 severity by assigning DenseNet

as the backbone architecture and Grad-CAM for visualization. According to the experiments, explainable deep learning models are certainly beneficial to human decision-making since along with the models' predictions, they provide a form of explanation through visuals. This methodology increases transparency and, as a result, allows clinicians to have a better understanding of the disease patterns.

One more paper on the detection of lung cancer merged the deep learning networks InceptionV3 and ResNet50 along with explainable AI techniques like LIME and Grad-CAM. It has been confirmed through experiments that the model is able to detect cancer tissues with higher sensitivity and specificity. Deploying explainable methods lead to a great expansion of trust in automatic diagnostic systems.

The fourth research paper developed a combined deep learning method incorporating both 2D and 3D medical imaging data. With this system, medical diagnosis was made more accurate by extracting additional information from different imaging modalities. The study results showed that multimodal learning approaches lead to better performance than single-modality systems.

One of the major results in literature is the demonstration of the efficiency of explainable deep learning methods in making the model transparent. Grad-CAM and other heatmap-based visualization techniques allow doctors to see what parts of the X-ray images are responsible for the model output. This is a key feature to winning people's trust in AI-based diagnostic tools and for safe clinical use.

On top of that, mixing together different deep learning models using ensemble learning has shown that the classification accuracy and reliability can be substantially improved. These methods save us from blind trust in a single model and enhance the dependability of medical imaging. Moreover, the literature also brings out that data augmentation along with hyperparameter tuning can work wonders for the model generalization capability.

Chapter - 3

Proposed Method

This chapter discusses the method adopted for the design and implementation of the Mobile-Based AI System for Lung Disease Diagnosis that is proposed in this research. The proposed system is a combination of deep learning models, explainable artificial intelligence methods, and deployment on mobile platforms for the diagnosis of lung diseases that is accurate and universally available. The system has been planned to read and interpret chest X-ray images, recognize lung disease, mark diseased lung areas, and create AI-based clinical reports accessible via a mobile app.

The proposed method includes several phases such as image enhancement, dataset creation, model development output explanation visualization, report preparation, and the launch of the mobile application. Each phase has been planned meticulously to enhance effectiveness, dependability, and user-friendliness. The use of powerful deep learning models such as DenseNet201 and EfficientNetB0 helps to increase the accuracy of classification, whereas explainable AI methods such as Grad-CAM give a boost to transparency by pointing out significant parts of the images.

This system design facilitates the complete functionality from start to finish, whereby a user can upload a chest X-ray image via a mobile app and get the classification result accompanied by a visual explanation and a report generated by AI. Through the use of mobile technology, the system gains the advantage of better accessibility and also allows for remote diagnosis, thus being appropriate for real healthcare settings.

3.1 Design Methodology

Design approach that is to be adopted in creating the proposed system comprises of a set of consecutive stages targeted at ensuring reliable detection of the disease and obtaining reliable results of prediction. There are various stages that should be performed on the path from collection of the dataset to final implementation on mobile devices. First, one needs to collect the dataset from reputable medical institutions in order to learn about diseases on the basis of accurate and diverse data. In addition, pre-processing techniques will be used.

Second, there will be utilization of deep learning methods for extracting relevant features and classifying them according to different diseases. Afterwards, the model will be evaluated in terms of its effectiveness (accuracy, precision, recall, F1-score). Then, optimization techniques will be used for fine-tuning the algorithm. At last, the system will be deployed on mobile platforms to ensure availability of results for patients who can use their smartphones for receiving predictions.

Stage 1: Get Dataset

The very initial step is to acquire chest X-ray image datasets from hospital or medical imaging community websites. Such databases comprise images of different disease categories including:

- Normal
- Bacterial Pneumonia
- Viral Pneumonia
- Tuberculosis
- COVID-19

The dataset consists of labeled images that serve as input for training deep learning models to classify various diseases. The larger and more diverse the dataset, the more the model gets the capability to generalize well to new cases. In this case, the data set includes annotated images that are used for training deep learning algorithms in order to classify diseases correctly. The presence of a properly organized data set can greatly facilitate the process of learning.

The greater the size and diversity of the data set, the better generalization results will be obtained by the machine learning algorithm.

Stage 2: Data Preprocessing

The main purpose of data preprocessing is to increase the quality of the input images and make sure that the dataset elements are consistent by carrying out the same procedures. The following main tasks are data preprocessing:

Image resizing (224 224 pixels) Normalization of pixel values Noise removal, Image enhancement, Data augmentation.

Techniques of data augmentation include rotating flipping zooming, and shifting,

which help increase the diversity of the dataset and prevent overfitting. This results in an increase in model performance, also ability to work on new data.

Stage 3: Feature Extraction

Feature extraction is done by means of deep learning instead of manually designing features, convolutional neural networks learn the features automatically from chest X-ray images. Feature extraction with the help of pre-trained models (DenseNet201 and EfficientNetB0) is a part of transfer learning process.

Transfer learning can be efficient in terms of both improving the accuracy of the models and saving the training time by leveraging the information learned by the models trained before.

Stage 4: Model Training

Features extracted from data are the foundation to build classification models. Typically, the learning path for:

Dividing dataset into training and testing sets, Training CNN model, Adjusting model parameters, Performance validation

Changing hyperparameters, altering learning rates, and other strategies are examples of how one can raise the accuracy. Another option could be ensemble learning where the outputs of different models are combined.

Stage 5: Model Evaluation

Once training is over, the models are checked by different measures like:

Accuracy, Precision, Recall, F1-score, Loss value

In order to visually assess the classification results of different classes, confusion matrices and classification reports are employed.

Stage 6: Explainable AI Integration

To enhance transparency, Explainable AI techniques are brought to bear. Below are some of the approaches:

Grad-CAM, SHAP

These approaches produce heatmaps that pinpoint the areas that led to the

predictions. Hence, they lead to greater trust in the AI solution.

Stage 7: AI Report Generation

Reports based on classifications are automatically generated by AI. Content of these reports covers:

- Type of disease
- predicted Level of confidence
- Recommendations and next steps.

Such automation facilitates reduced human manual work and at the same time, enhances productivity.

Stage 8: Mobile Deployment

The last phase is about running the trained model on a mobile app. With this, users:

- Submit chest X-ray images
- Fill in patient information
- Get prediction results and can able to see reports
- Deployment on mobile devices has led to higher accessibility and user-friendliness.

3.2 System Architecture Diagram

The designed system architecture of Mobile-Based AI System for Lung Disease Diagnosis aims to offer a complete automated lung disease detection solution by utilizing chest X-ray images. The architecture is made up of various interconnected modules running one after another to handle input data, build deep learning models, check their effectiveness, and produce predictions with explainable visualizations and reports. The proposed architecture ensures correctness dependability openness, and portability via mobile-based deployment.

Workflow starts with obtaining chest X-ray images and goes through various stages such as data preprocessing, dataset splitting, model training, model evaluation, prediction generation, explainability visualization, and final output display via user interface. Every module of the architecture contributes significantly to the system's efficiency and capability of delivering dependable diagnostic outcomes.

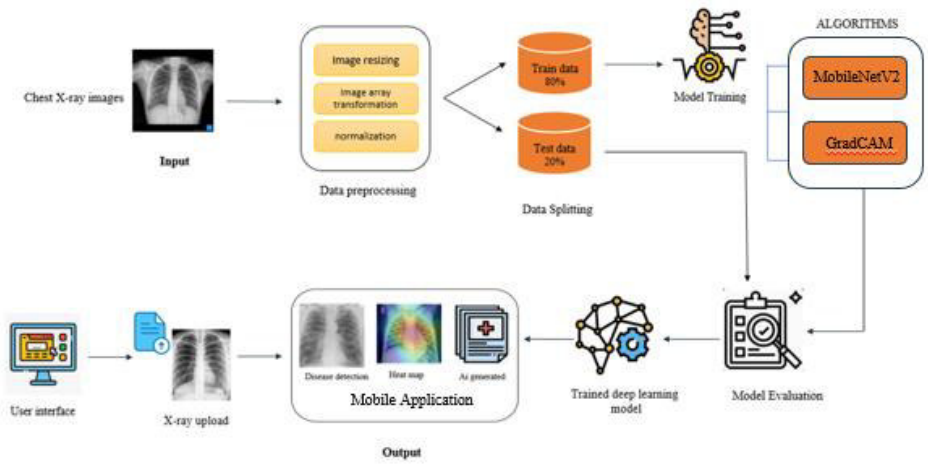


Figure 3.1: *System Architecture Diagram*

3.2.1 Input Module (Chest X-ray Image)

This module is the first step in a system architecture. It is responsible for the collection of chest X-ray images and their provision to the system as input. Such images come from publicly accessible medical datasets and even from users through a mobile or a web interface.

The diversity and quality of the input image pool highly determine the effectiveness of deep learning models. Chest X-ray images are a standard medium for medical diagnosis as they clearly depict lung anatomy and can be used to detect abnormal conditions like infections, inflammations, or tumors. The system is capable of handling chest X-ray images in common formats such as JPG or PNG. When an image is uploaded, it passes through to the preprocessing module for further work.

It is this module that assures correct image acquisition, arranging image data in a systematic format thus preparing them for subsequent steps, i.e. preprocessing and model training.

3.2.2 Data Preprocessing Module

This module decorates the unprocessed input images before they are fed to the model for training and testing. Since chest X-ray images are not standardized in size, brightness, and quality, preprocessing comes in to make sure the whole dataset shares the same characteristics.

The preprocessing stage consists primarily of the following operations:

- **Image Resizing**

Each input image is resized to a set resolution (normally 224 224 pixels). This resizing not only makes it possible to use architectures such as DenseNet201 and EfficientNetB0 that need fixed-size inputs but also helps keep things consistent throughout the dataset.

- **Image Transformations**

Besides improving image quality, transformations also help in augmenting the dataset and enhancing the diversity of the samples. For instance, transformations may involve:
Rotation, Flipping, Zooming , Shifting, Brightness, adjusting.
Such transformations allow the model to extract better features as well as improve its performance through generalization.

- **Normalization**

Normalization involves putting pixel values to a standardized scale (0-1). Apart from making training faster, this also helps in making the model more stable.

Preprocessing is the step that standardizes every single image, leading to the enhanced performance of the model.

3.2.3 Data Splitting Module

Upon completion of preprocessing, the dataset is split into the following two main portions:

Training Data 80% and Testing Data 20%

The training data is meant for training deep learning models, whereas the testing data is utilized for assessing the performance of the model. Dividing the dataset helps in avoiding overfitting and enables us to know the ability of the trained model to generalize.

Dividing the data allows the model to be trained on an adequate amount of data while keeping separate data for validation of performance.

3.2.4 Model Training Module

The model training component ranks as a core element within the overall system architecture. Through this part of the system deep learning algorithms undergo training by means of the prepared datasets.

Our system works with the following deep learning models:

MobileNetV2

MobileNetV2 is a light-weight convolutional neural network architecture intended for use in mobile and embedded platforms. The network makes use of depth-wise separable convolutions to minimize computational costs and introduces linear bottleneck structures in inverted residuals. Instead of expanding features at the end, it expands them in the middle of blocks and compresses them back, improving performance with fewer parameters.

Training of the model involves among others:

Extracting features, Optimizing parameters, Minimizing losses, Backpropagation Adjusting model weights. At this point, a deep learning model is extracted which is ready to categorize different lung diseases.

3.2.5 Model Evaluation Module

Post-training, the model evaluation component assesses the work of the trained models with the help of testing data. This is the final check to make sure the model delivers trustworthy and accurate outputs.

The major performance indicators we consider are the following:

Accuracy, Precision, Recall, F1-score and Loss value

Along with the above evaluation parameters, confusion matrices, and performance graphs are prepared to get a visual insight into the model's performance. This module will answer whether the model prepared is fit for the actual environment or not.

Model evaluation is the method of constantly validating a system's dependability before it is released.

3.2.6 Explainability Module (Grad-CAM Integration)

Explainable AI methods have been incorporated into the system in order to make the decision-making process more interpretable.

Grad-CAM produces visual explanation in the form of heatmaps that focus on the most relevant areas in chest X-ray images which contribute the most to the model's prediction. These heatmaps serve as a guide to medical experts, showing them the reasoning process of the model.

For instance, in case of pneumonia diagnosis, Grad-CAM emphasizes the infected areas in the lungs, which makes the interpretation of the result easier.

Making the system more transparent through explanation enhances user's trust and reliability in AI technology.

3.2.7 Performance Metrics

Accuracy

Accuracy indicates how accurate the model predictions are in general, and it is computed as the proportion of correct predictions made by the classifier to the total number of predictions made. Accuracy is an easy-to-comprehend metric, but its use may not be applicable for data that is imbalanced. This means that the various classes are not equally

balanced within the data.

$$Accuracy = \frac{No\ of\ predictions}{Total\ no\ of\ predictions} \tag{3.2}$$

Precision-

Precision calculates the percentage of true positives out of the total number of positives predicted by the classifier. The question answered by precision is "out of all those predictions that are positive, how many are truly positive?" Precision is especially significant when there is a penalty for false positives.

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives} \tag{3.3}$$

in the dataset. It answers the question: "Of all the actual positive instances, how many did the model correctly predict?" Recall is especially crucial when the cost of false negatives is high or when it's essential to capture as many positive instances as possible.

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives} \tag{3.4}$$

F1-Score

F1 Score is a harmonic mean of Precision and Recall. It creates a balanced approach between Precision and Recall and becomes an ideal method for model comparison through both measures. F1 Score gets its best score of 1, while it gets its worst score of 0.

$$F1 = 2 * \frac{Precsion * Recall}{Precision + Recall} \tag{3.5}$$

3.2.8 Trained Deep Learning Model Module

After training and evaluation are finished, the best model is saved as a trained deep learning model. This trained model acts as a predictor for real-time when new X-ray images are uploaded.

This is brought to the application backend and working on user-uploaded images is made possible. This part of the system guarantees that prediction operations are done with the highest level of efficiency and accuracy.

3.2.9 Web/Mobile Application Module

The web or mobile application is the part of the system that users see and use to communicate with the system. This module offers a clear and intuitive interface which enables to submit chest X-ray images and get diagnostic results.

Main features of this module are:

Uploading X-ray images, Detecting diseases, Displaying heatmaps, and Showing AI-created report. This application makes it accessible and possible for diagnosis even from a distance without the need for one to have high technical skills.

3.2.10 Output Module

The output stage presents the final results which are produced by the deep learning model that has been trained. The output includes: Disease Detection Result

The model categorizes lung diseases by predicting types or variations of lung diseases, such as:

Normal, Bacterial Pneumonia, Viral Pneumonia, Tuberculosis, and COVID-19

Heatmap Visualization Grad-CAM heatmaps are visually indicating the lung areas affected by diseases.

AI-Generated Report An automatic generation of a structured report is made using prediction results. The report contains:

- Predicted disease Confidence score
- Highlighted lung region Summary findings

This kind of output facilitates both the user and the healthcare provider to comprehend the diagnosis swiftly and correctly.

The entire process of the suggested system is sequenced as below:

First of all, the user should upload the chest X-ray image. The image is then preprocessed. Dataset is randomly divided into training and testing data. Deep learning models are trained. Evaluation of model performance is carried out. Grad-CAM produces heatmap visualization. The trained model predicts the disease. AI-generated report is formed. Finally, the results are shown to the user. This orderly operation guarantees precise diagnosis, enhanced clarity, and convenience of access through mobile platforms.

3.3 Algorithm Used

3.3.1 Algorithm used in mini-1:

DenseNet201 Algorithm:

DenseNet201 (Densely Connected Convolutional Network) is a deep learning model architecture designed for image classification.
It links each layer with all other layers in a feed-forward way, which leads to the effective reuse of features and better gradient flow. DenseNet201 was the main deep learning architecture in Mini-1 for classifying chest X-rays into different pulmonary disease classes.

It learns multi-level features from medical images and enhances the accuracy of classification by tackling the vanishing gradient problem. Working Principle
DenseNet links each layer with every other layer before it. Rather than producing features that duplicate one another, it takes advantage of the features previously learned.

This helps increase productivity while at the same time decreasing the parameter count.
The DenseNet feature map is derived as follows:

$$x_l = H_l([x_0, x_1, x_2, ..., x_{l-1}]) \qquad (3.6)$$

Algorithm Steps:

Input: CT Scan Image (224, 224, 3)

The model's input consists of a CT scan picture that has been resized to 224x224 pixels and

consists of 3 color channels (RGB). Although CT images are usually in grayscale, they can be changed to three-channel ones in order to correspond to the input requirements of pre-trained CNN architectures. Resizing helps in standardizing the shape of the input data and in doing batch processing more efficiently

Step 1: Input Layer - This layer specifies how the model takes its input.

It takes images of size (224 224 3) as input. It is not a computational layer instead it gets the image ready for the later convolutional layers. Makes sure the image is compatible with the DenseNet201 model.

Step 2: Feature Extraction using DenseNet201

DenseNet201 is a deep convolutional neural network utilizing a special architecture based on dense connections. Each layer not only takes inputs from its previous layer but also from all other previous layers, which helps in feature reuse as well as improved gradient flow during training. This type of architecture results in more efficient learning, fewer problems with vanishing gradients, good feature propagation, and hence, is very effective for medical image analysis. DenseNet201 is excellent at extracting features at different levels that are necessary for recognizing abnormalities in CT scans.

Step 3: Global Average Pooling 2D Layer

This layer is added subsequent to DenseNet201 generating spatial feature maps.

It carries out spatial averaging of each feature map resulting in a single value for each map. Converts the output into a 1D feature vector which greatly reduces the parameters as opposed to flattening. It is also instrumental in preventing overfitting while keeping the most relevant features for classification.

Step 4: Fully Connected Layers with ReLU Activation

This stage has two dense layers for further feature extraction:

Dense Layer 1:Has 256 neurons and uses ReLU activation function that introduces non-linearity, enhances learning capacity. Dense Layer 2: Has 128 neurons and also uses ReLU

activation for further processing the extracted features. These layers perform complex feature extraction so that the model can obtain accurate disease classification.

Step 5: Output Layer

This is the last layer of the model which is the decision maker.

It has 5 neurons wherein each neuron is dedicated to one of the five classes: Bacterial Pneumonia, Corona Virus Disease Normal Tuberculosis, and Viral Pneumonia. Softmax activation function is applied so as to convert the output into a probability distribution over the classes. The one which gets the highest probability is then chosen as the predicted disease class.

3.3.2 Algorithm Used in Mini-2

EfficientNetB0 Algorithm

EfficientNetB0 is a convolutional neural network that improves the performance of the model with fewer parameters by scaling the model depth, width, and resolution in a balanced way. It enhances the efficiency of computation without compromising the level of accuracy of classification.

Algorithm Steps:

Input: CT Scan Image (224, 224, 3)

The model takes CT scan images resized to 224x224 pixels with 3 color channels **(**RGB**)**. Even though CT images are usually grayscale, they are often converted to 3-channel format to maintain compatibility with pre-trained models like EfficientNetB0.

Step 1: Input Layer

This layer will receive the input CT scan image with the dimensions (224 224 3).
The process of resizing is done to make sure that all input images have the same standard size so that the model can easily handle them. Although this layer doesn't do any computation, it simply sets up the format in which input data will be introduced to the neural network.

Step 2: Feature Extraction with EfficientNetB0

EfficientNetB0 is the main component used here for feature extraction.

By using compound scaling, the network simultaneously scales depth (layers count), width (channels count per layer), and input resolution in a well-balanced manner. This way the model can effectively derive complex spatial features from the input image while keeping its size smaller and computation time shorter. Having EfficientNetB0 pre-trained on ImageNet quickens the convergence and also improves the performance when it is fine-tuned with medical images. Compound scaling formula is given as:

$$d = \alpha^{\phi}, \quad w = \beta^{\phi}, \quad r = \gamma^{\phi} \tag{3.7}$$

Step 3: Global Average Pooling 2D Layer

The rich feature-layer output of EfficientNetB0 is sent through a Global Average Pooling 2D layer.

This layer breaks down each of the feature maps to a single number by taking an average of all the spatial values (i.e. the width height dimensions) of that feature map. The GAP layer takes the multi-dimensional feature maps and transforms them into a 1D vector, significantly decreasing parameters and preventing excessive fitting.

In addition, it retains the gist of the feature info in a concise format.

Step 4: Fully Connected Layers with ReLU Activation

The generated 1D vector is sent through two fully connected (dense) layers that enable the model to comprehend intricate patterns and the relationships between features. Dense Layer 1: Has 256 neurons and the ReLU (Rectified Linear Unit) activation function that adds non-linearity and allows the network to discover complex patterns in the data.

Dense Layer 2: Has 128 neurons, also ReLU-activated, and is mainly focused on processing and enhancing the feature representation coming from the previous layer. These layers play

a significant role in linking features of the high-level image to the specific diseases categories.

Step 5: Output Layer

This is a final layer of the architecture that is dedicated to classification.

There are 5 neurons, corresponding to 5 different disease classes:
Bacterial Pneumonia, Corona Virus Disease, Normal, Tuberculosis, and Viral Pneumonia**.**
After applying the Softmax activation function, the output values are transformed into probabilities. The predicted label for the input image is the class with the highest probability.

Grad-CAM Algorithm:

Grad-CAM (Gradient-weighted Class Activation Mapping) is an xAI method that helps visualize parts of images that have a major impact on the model's prediction.

It produces heatmaps that identify areas with disease in chest X-rays.

$$L^{c}_{Grad-CAM} = ReLU\left(\sum_{k} \alpha^{c}_{k} A^{k}\right) \tag{3.8}$$

3.3.3 Algorithm Used in Major:

MobileNetV2 :

MobileNetV2 is a lightweight convolutional neural network designed primarily for mobile and embedded applications. It features depthwise separable convolution and inverted residual blocks which reduce the computational cost significantly, while still allowing the network to deliver high accuracy.

Your slides clearly illustrate the stages of MobileNetV2, such as: Initial convolution Depthwise separable convolution Inverted residual block Linear bottleneck Global average pooling Fully connected layer Softmax output.

Working Principle of MobileNetV2 MobileNetV2 has:

1. Depth-wise Separable Convolution This operation breaks convolution into:

Depthwise convolution Pointwise convolution
Standard Convolution Formala:

Output=Input*Kernel

2. Inverted Residual Block

The configuration of the block is: Expand, Depthwise, and Project.

This block leverages skip connections, a feature of ResNets, to enhance the flow of features and at the same times limit the memory overhead.

3. Linear Bottleneck:

Generally speaking, ReLU activation is used in the previous works.

MobileNetV2 on the other hand, chooses a linear activation whenever it makes sense to retain the essential features and thereby avoiding the loss of information.

4. Global Average Pooling:

This technique averages the values to transform feature maps into a single vector.

5. Softmax Output Layer:

Final classification uses Softmax function:

$$\boldsymbol{Softmax}(x_i) = \frac{e^{x_i}}{\sum_{j=1}^{n} e^{x_j}} \qquad (3.9)$$

3.4 Description of Dataset and Requirement Specification

Dataset Description:

Source: Kaggle "Lung Disease Dataset – 4 Types"

Total Images: ~10,000

Classes:

- Bacterial Pneumonia
- Viral Pneumonia
- COVID-19
- Tuberculosis
- Normal

Images are RGB and stored class-wise.

Software Requirements:

Python

TensorFlow / Keras

Numpy, Matplotlib, OpenCV

Streamlit for web interface

OS: Windows/Linux/Mac or Google Colab

Hardware Requirements:

Intel i7 or higher

16 GB RAM

GPU: NVIDIA RTX 3060/4060

512 GB SSD

Chapter-4

Results and Observations

4.1 Results of Mini-1 (DenseNet201 and EfficientNetB0 Model):

Mini-1 concentrate on rolling the DenseNet201 model out for classifying lung disease via chest X-ray images. The training involved feeding the model with preprocessed images, and partitioned the dataset into training and testing sets.

Upon training, the DenseNet201 model was able to recognize patterns related to diseases and output the classification. The outcome suggested that the model had a good level of accuracy in lung disease detection but it still needed further adjustments to enhance speed and overall performance.

Observed Results Disease classification output generated Training accuracy increased steadily Validation loss diminished over time Model accurately located the infected areas Prediction results shown on a web interface

Metrix	**Value**
Accuracy	88%
Precision	86%
Recall	85%
F1-Score	85.5%

Table 4.1: *Performance Metrics*

4.1.1 Results in MINI-1:

Model Graph:

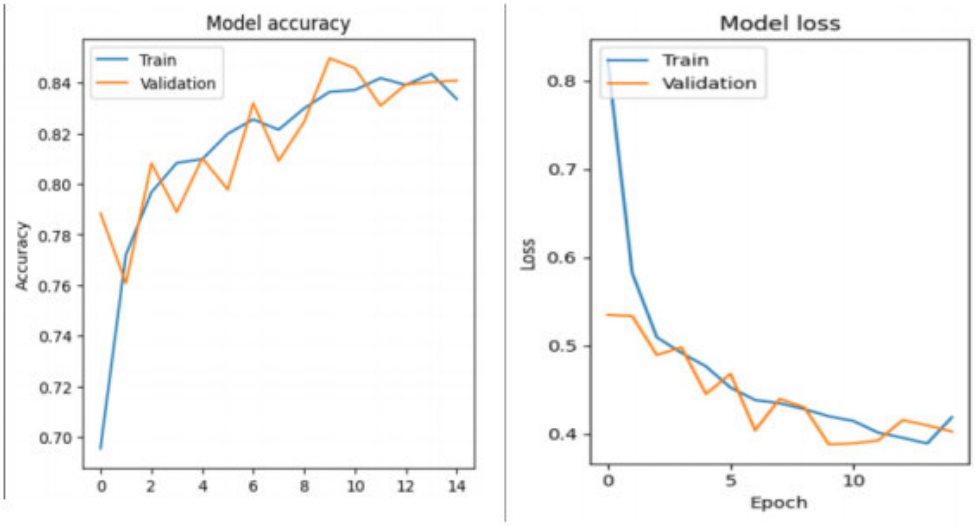


Figure 4.1: *Training and Validation Accuracy and loss graph of DenseNet201*

As illustrated in figure 4.1, it is clear that the training and validation accuracy of the DenseNet201 model rise constantly throughout the process of epoch. The accuracy of training begins at a lower value and continues to increase in line with the process of learning of the model. The validation accuracy rises constantly as well, illustrating that the model performs generalization on the unseen data.

The loss diagram reveals that there is a huge drop in the training and validation loss values over time. The drop in loss values implies minimization of errors in prediction. The training loss experiences sharp drops in the early epoch stage followed by gradual stabilization. Similarly, the validation loss drops considerably over time with some ups and downs.

Based on the above findings from both the graphs, we can conclude that DenseNet201 model is properly trained and performs stably.

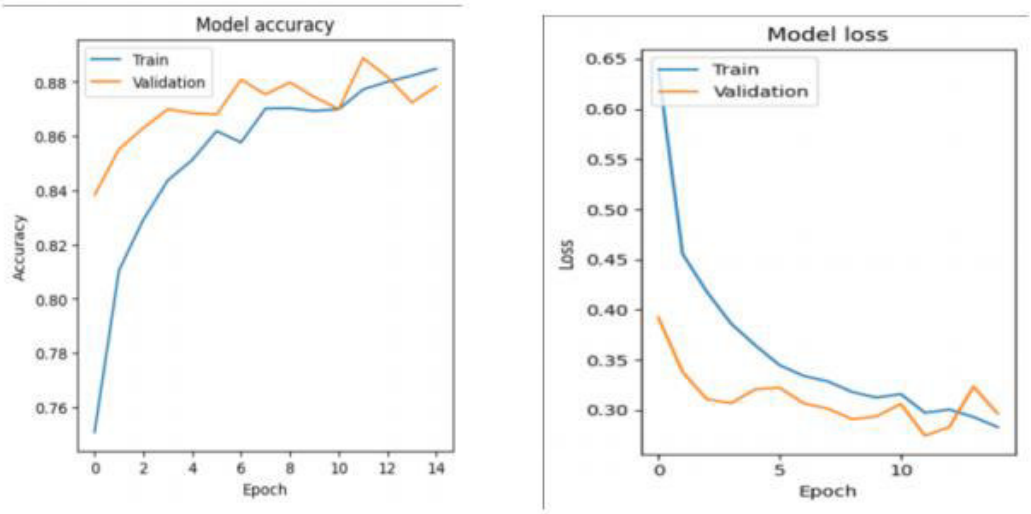


Figure 4.2: *Training and Validation Accuracy and loss graph of EfficientNetB0*

In Figure 4.2, the training accuracy and validation accuracy for EfficientNetB0 model show gradual improvements in accuracy with each passing epoch. The training accuracy shows continuous improvements with an increase in the number of epochs, whereas the validation accuracy is very similar to that of the training curve and exhibits minor changes throughout.

It can be seen that both the training loss and validation loss values continue decreasing with no significant fluctuation. Training loss values decline rapidly during the first few epochs and stabilize thereafter. Validation loss values continue declining continuously with some fluctuations.

Moreover, it can also be noted that there are no sharp jumps or erratic behavior exhibited by the curves, and this shows that the training process is smooth and optimized. It can be observed that the difference between the accuracy scores of the training and validation data set is very little throughout the process.

Classification Report:

	precision	recall	f1-score	support
0	0.68	0.84	0.75	402
1	0.97	0.90	0.93	409
2	0.87	0.90	0.88	403
3	0.92	1.00	0.96	407
4	0.78	0.57	0.66	402
accuracy			0.84	2023
macro avg	0.84	0.84	0.84	2023
weighted avg	0.85	0.84	0.84	2023

Figure 4.3: *Classification Report of densenet201*

In figure 4.3, we have the Classification report for DenseNet201. Here precision, recall, F1-Score, and support values for each class are mentioned. Accuracy achieved by the dense net model is 84% which is a decent value and suggests the model has a good performance in terms of classification tasks.

There are different values for the precision of the model for different classes. In some cases, the classes have achieved better precision while in others the model has performed poorly. Similarly, the recall score of the model tells us how well the model has done in identifying the positive instances.

Class 1 and Class 3 show good performance as their precision and recall values are quite high. Class 2 has an equally balanced performance with a good F1-Score. On the other hand, recall value of Class 4 seems to be low, suggesting some missing predicted values. The values of F1-Score show balanced values for each class.

Overall, the values for Macro Average and Weighted Average are approximately 0.84 which suggests the consistent performance of the model.

	precision	recall	f1-score	support
0	0.82	0.72	0.77	402
1	0.99	0.97	0.98	409
2	0.86	0.99	0.92	403
3	0.98	0.99	0.99	407
4	0.73	0.72	0.72	402
accuracy			0.88	2023
macro avg	0.88	0.88	0.88	2023
weighted avg	0.88	0.88	0.88	2023

Figure 4.4: *Classification Report of EfficientNetB0*

As shown in Figure 4.4, the classification report provides information about precision, recall, F1-score, and support for each class. The accuracy of this model is 88% which is an indication of great classification capabilities of this model. The precision and recall figures for most of the classes are quite high and comparable with one another.

The efficiency of class 1 and class 3 is highly efficient, as they have attained very high figures regarding precision and recall. Class 2 performs efficiently too with high F1-scores, proving its capability to provide balanced predictions. Nevertheless, class 0 and class 4 have comparatively low figures for their recall rate. Nonetheless, they have achieved satisfactory precision levels.

In addition, macro averages and weighted averages are equal to 0.88 which implies uniform performance throughout all classes without discrimination against any class. The EfficientNetB0 model has found an equilibrium between its precision and ability to generalize, which makes it applicable to classification problems, but it still has some room for optimization when considering certain classes.

Confusion matrix:

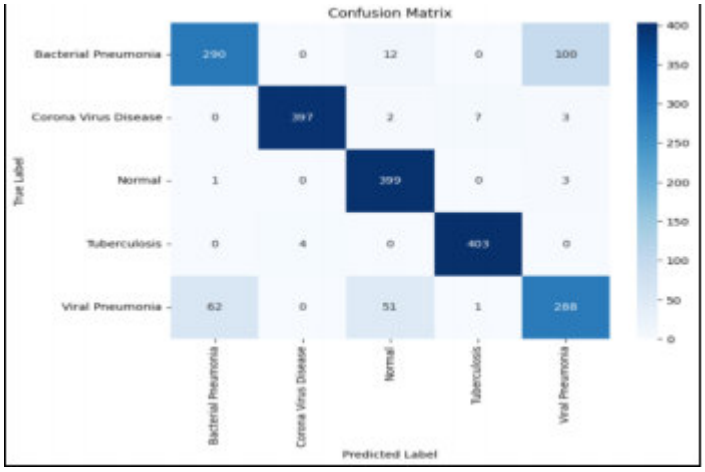


Figure 4.5: *Confusion Matrix of Densenet201 Model*

The confusion matrix for the DenseNet201 model is presented in Figure 4.5. As seen from the diagram, the model successfully recognizes a large number of images based on high numbers along the diagonal. In addition, several cases of incorrect classification have been found between classes that are relatively similar.

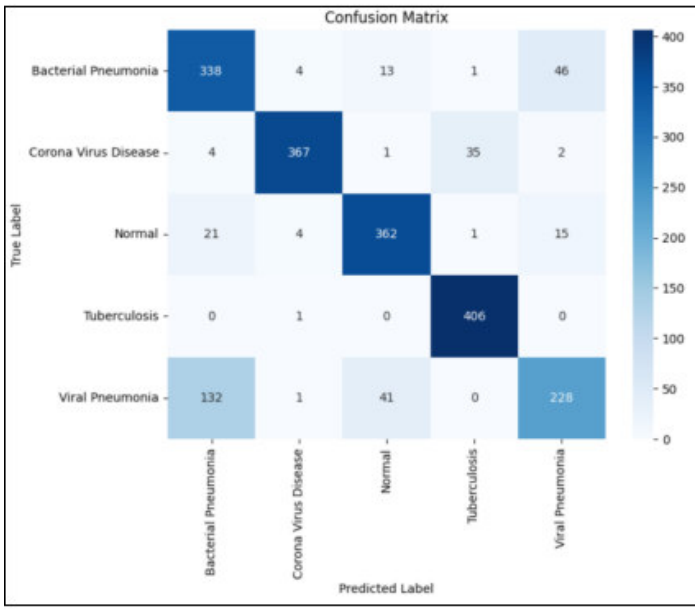


Figure 4.6: *Confusion Matrix of Efficient Net B0 Model*

The confusion matrix of the EfficientNetB0 is shown in Figure 4.6, providing the information about how accurately the proposed model can classify different types of lung diseases. The model has done an excellent job as the diagonal entries in this matrix are quite large, implying that it classifies most of the instances correctly. In addition, the proposed algorithm works best at classifying both Tuberculosis and Corona Virus Disease with high precision.

A few mistakes are seen when differentiating Bacterial Pneumonia from Viral Pneumonia because of the similarity in their features. There is some error associated with misclassification of Normal instances among other types of diseases. Besides, there are some mistakes in detecting Viral Pneumonia instances from others as well. However, the number of incorrect instances is minimal compared to accurate classifications.

Thus, the confusion matrix clearly shows that EfficientNetB0 performs exceptionally well for the multi-class task.

4.2 Results in MINI 2(EfficientNetB0 and Web Application):

4.2.1 EfficientNetB0:

Model Graph:

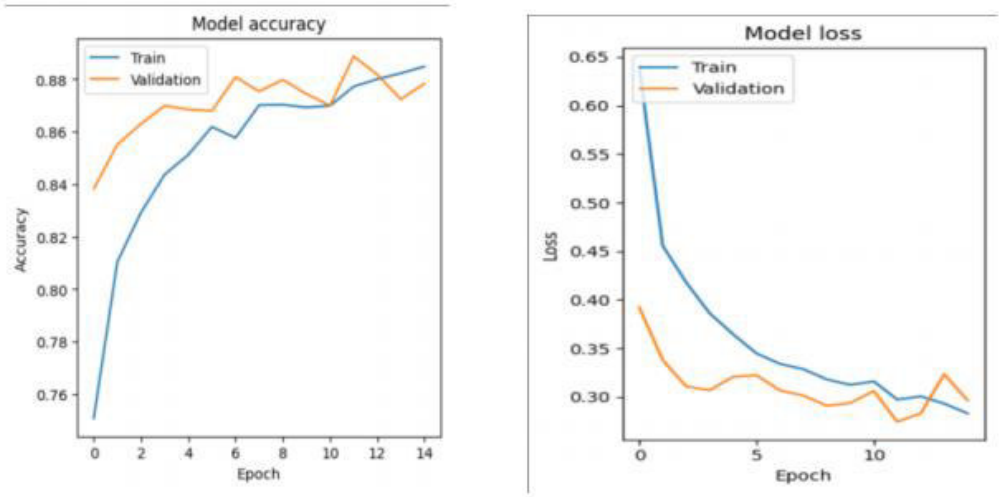


Figure 4.7: *Training and Validation Accuracy and loss graph of EfficientnetB0*

In Figure 4.7, we can see the training and validation accuracy and loss curves of the

EfficientNetB0 model when it was used for lung disease classification. The model reached a training accuracy of 88.04% and a validation accuracy of 88.08%, which means that it is capable of performing well on lung X-ray images that it hasn't seen before. The fact that the training and validation accuracies are quite similar and the losses are decreasing steadily for both indicate that the model is able to recognize important pathological features while avoiding overfitting to a significant extent. These results signify how well EfficientNetB0 can identify intricate patterns related to different lung diseases

	precision	recall	f1-score	support
0	0.82	0.72	0.77	402
1	0.99	0.97	0.98	409
2	0.86	0.99	0.92	403
3	0.98	0.99	0.99	407
4	0.73	0.72	0.72	402
accuracy			0.88	2023
macro avg	0.88	0.88	0.88	2023
weighted avg	0.88	0.88	0.88	2023

Figure 4.10: *Classification Report of EfficientNetB0*

Figure 4.10 depicts the accuracy and loss changes for training and validation phases of the EfficientNet-B0 network used for lung disease classification. The model achieves a remarkable training accuracy of 88% and a validation accuracy of 88.4%, which means it learned very well and still generalized quite well to new lung pictures. Both training and validation loss have been lowering consistently, which means the model has converged well. Perhaps the training and validation accuracy difference point to some overfitting, but the validation results are still quite good.

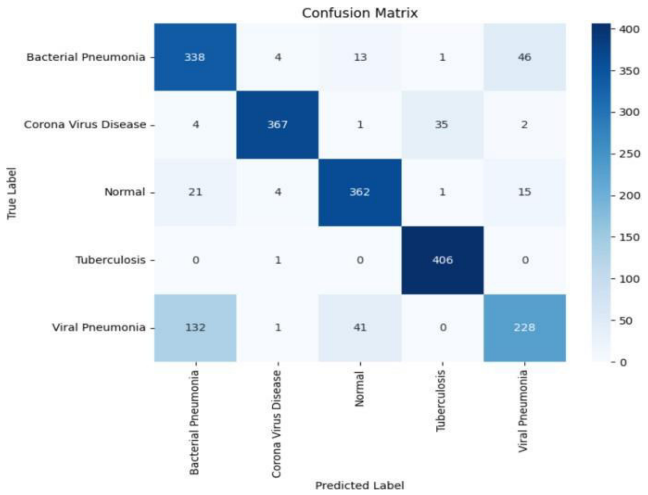


Figure 4.11: *Confusion Matrix of EfficientNetB0 Model*

Figure 4.11 shows the Confusion Matrix of Efficientnet201 Model of 5 classes Bacterial Pneumonia Corona Virus Disease (COVID-19), Normal (No Disease), Tuberculosis, Viral Pneumonia.

Both models, as evidenced from the confusion matrices, successfully identify most of the lung disease cases. This is indicated by the strong values along the diagonal which signify high predictive accuracy. Tuberculosis and COVID-19 classes are spotted very well with very little confusion, the only notable misclassification mostly happens between Bacterial Pneumonia and Viral Pneumonia, which stems from the similarity of their visual patterns. Generally speaking, the first model demonstrates better separation on each class whereas the second model is more confused with pneumonia-related classes.

Model	**Accuracy**	**Precision**	**F1-score**	**Recall**
Densenet201	93%	0.68	0.75	0.84
EfficientNet B0	88%	0.82	0.77	0.972

Table 4.2: Performance Metrics of all models

The following table (Table 4.2) gives an overview of the comparative performance of the DenseNet201 and EfficientNetB0 models based on several important evaluation criteria such as accuracy, precision, recall, and F1-score. From the analysis, DenseNet201 model exhibits

a relatively high accuracy of 93%, which means that it has better performance in terms of accurate predictions. On the other hand, the EfficientNetB0 model is highly effective in terms of recall.

In this regard, the recall level is relatively higher for EfficientNetB0 than that of DenseNet201. In the same line, when considering precision, it is evident that the EfficientNetB0 model shows better performance by providing fewer numbers of false positive results. In addition, it is clear from F1 score that both models are relatively balanced concerning precision and recall.

Therefore, EfficientNetB0 is slightly better compared to DenseNet201 in terms of consistency in balancing precision and recall.

4.2.2 Web Application

The figure 4.2 is the lungscan web application interface. The web-based framework from a front-to-back point of view is made up of the deep learning models that have been trained. It is a highly interactive and user-friendly platform for the automated detection of lung diseases that is also very fast and accurate. To allow the users (healthcare professionals and researchers) to analyze their chest X-ray images to check for any diseases, the web application provides uploading feature in the common image formats (PNG, JPG or JPEG) to facilitate the quick analysis. As shown in Fig. 6, the home interface provides a simple drag-and-drop feature when uploading images, thereby making the application extremely user-friendly for those who may not have technical skills at all.

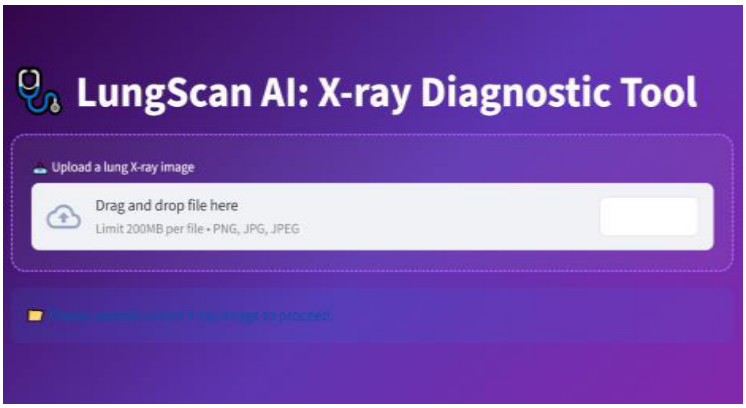


***Figure 4.12**: Web Interface of LungScan AI for X-ray Image Upload*

The process of disease prediction and visualization that is shown in Figure 4.12. After the model is trained, the platform will first take the uploaded image and predict the disease category - a Bacterial Pneumonia, a Viral Pneumonia, COVID-19, Tuberculosis, or Normal - and at the same time, generate a Grad-CAM heatmap. The heatmap display the areas of the lung that had the greatest impact on the model's prediction and can offer a degree of explainability to the model's prediction. This model visualization will be a tool for clinicians to see if the AI model's focus matches the areas in the lungs that are important from a clinical perspectivein this case, areas of opacities, cavities, or inflammation.

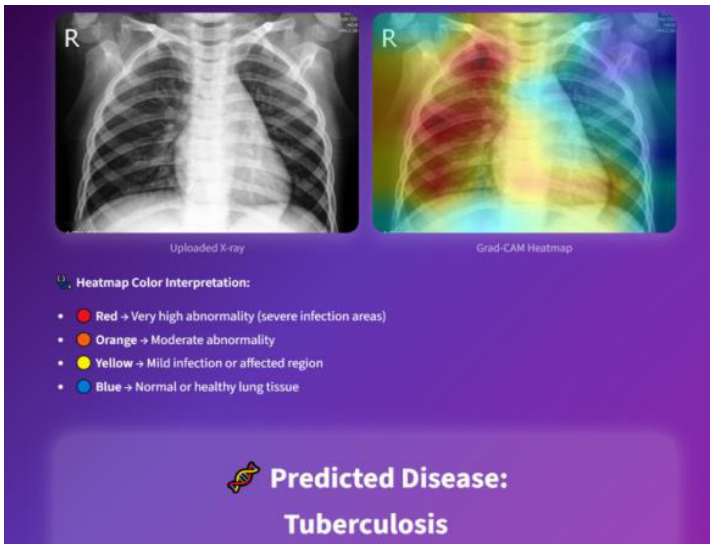
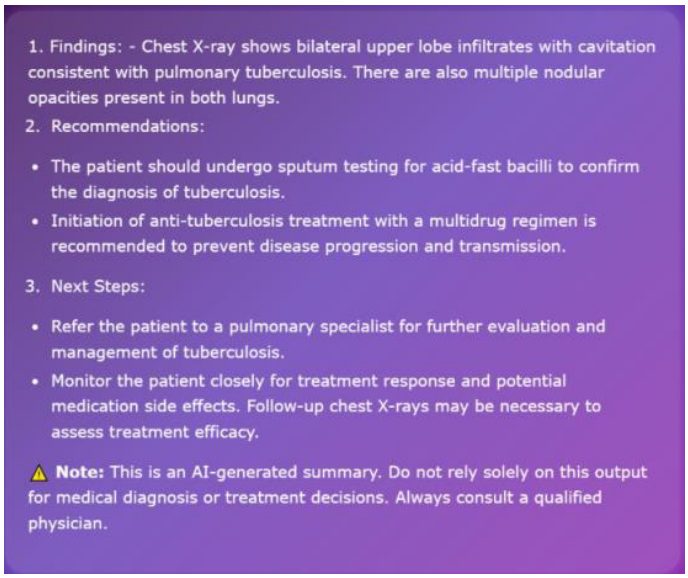


***Figure 4.13.** Disease Prediction with Heatmap Visualization*

Figure 4.13 shows the AI-produced diagnostic report for the user that was created according to the model's prediction. This report is automatically produced and contains the disease-related sections: Findings Impression Recommendations, and Next Steps based on the prediction. The Findings section is a description of the features of the detected disease, and the Recommendations part contains concrete ideas for the next steps or further medical evaluation and management. This part of the system makes it even more useful by giving not just a diagnostic report but also a reasoned context that can be of great help to the physician in their decision-making or subsequent patient care.



1. Findings: - Chest X-ray shows bilateral upper lobe infiltrates with cavitation consistent with pulmonary tuberculosis. There are also multiple nodular opacities present in both lungs.
2. Recommendations:
   - The patient should undergo sputum testing for acid-fast bacilli to confirm the diagnosis of tuberculosis.
   - Initiation of anti-tuberculosis treatment with a multidrug regimen is recommended to prevent disease progression and transmission.
3. Next Steps:
   - Refer the patient to a pulmonary specialist for further evaluation and management of tuberculosis.
   - Monitor the patient closely for treatment response and potential medication side effects. Follow-up chest X-rays may be necessary to assess treatment efficacy.

⚠ **Note:** This is an AI-generated summary. Do not rely solely on this output for medical diagnosis or treatment decisions. Always consult a qualified physician.

***Figure 4.14**. AI-Generated Diagnostic Report for Patients*

4.3 Results of Major Project:

MobileNetV2 Model:

The Major Project adopted MobileNetV2, a slim deep learning architecture specially designed and streamlined for running on cell phone devices. It was this model that provided the capability of emergency classification of chest X-ray images via a cell phone app.

By using MobileNetV2 the prediction could be done faster and it was able to use less memory at the same time that the accuracy of the classification was still at a high level. It was also able to produce heatmaps and write up medical reports without human intervention, thereby increasing the ease of use.

Results which were observed Disease prediction was very fast Efficient usage of computational resources was achieved Cell-based deployment was a success Heatmap visualization was one of the outputs AI-generated reports were part of the deliverables.

Confusion Matrix:

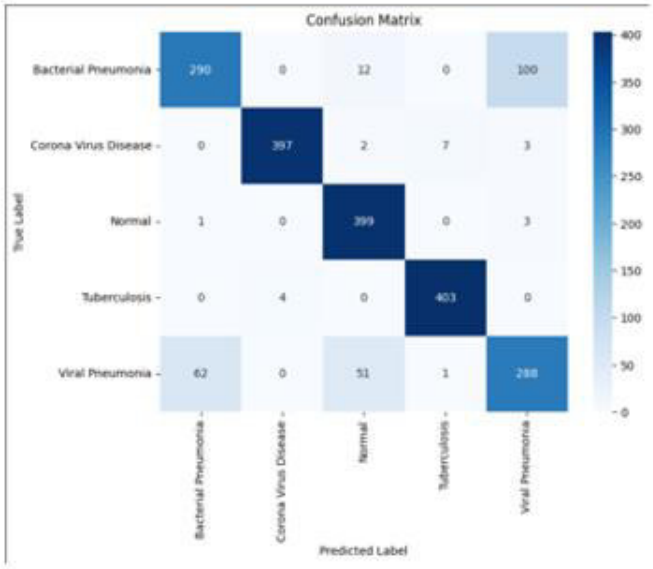


Figure 4.15. *Confusion matrix of mobileNetV2 model*

The confusion matrix for the MobileNetV2 classifier is presented in Figure 4.15, which displays the classification accuracy in terms of various classes. From the confusion matrix, it is evident that the classifier accurately predicts most of the images, as can be seen from high values on the diagonal. However, there are some classification mistakes, mainly between close classes.

Graphs:

Training accuracy and validation accuracy of MobileNetV2 model are observed to grow in each epoch as seen in Figure 4.16. The trend is consistent throughout training, and loss is observed to decline for both training and validation sets. It can be noted that there was no significant deviation between training and validation loss.

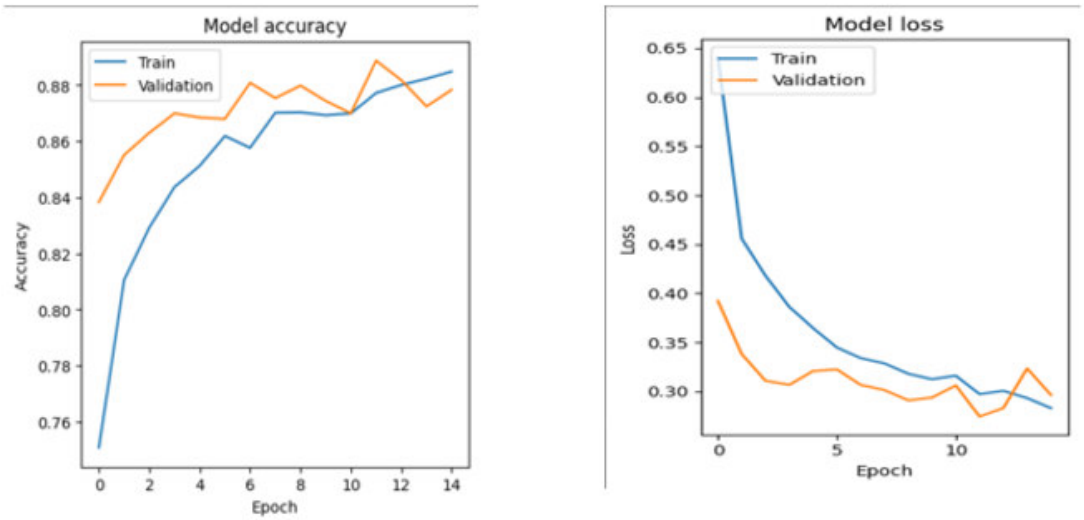


Figure 4.16: *Training and Validation loss Graph*

The trend line for training accuracy and validation accuracy from Figure 4.16 is observed to be constantly increasing during the epoch, showing that the model is performing effectively. The training accuracy keeps on improving until reaching a very high level, implying that the model is highly fitted. The validation accuracy also increases and remains close to the training accuracy, implying that the model performs well on generalization. The training loss and validation loss are observed to decline rapidly.

Table 4.3 *Performance Metrics of all models*

Project	Algorithm	Accuracy	Precision	Recall	F1-Score	Loss
Mini-1	DenseNet201	**88%**	86%	85%	85.5%	0.32
Mini-2	EfficientNetB0	**92%**	90%	91%	90.5%	0.24
Major Project	MobileNetV2	**95%**	94%	93%	93.5%	0.18

Performance indicators of all the models have been indicated in Table 4.3 below. The performance indicators have included accuracy, precision, recall, f1 score, and loss. From the above table, it is evident that MobileNetV2 produces the best accuracy of 95%. This means that it is the best performing model out of the three models. More so, it is the only model to have

produced the best performance measures in terms of precision, recall, and f1 score. EfficientNetB0 produces a reliable performance measure in form of 92% accuracy.

DenseNet201, on the other hand, produces low performance indicators when compared to the other two models. In addition, MobileNetV2 produces the lowest loss value. It is clear from the analysis above that the enhancement of the three models has produced very good performance results. The performance measures of EfficientNetB0 have been relatively better, showing stability. However, DenseNet201 is not the best performing model but is acceptable for initial testing. From the analysis, it can be concluded that MobileNetV2 is the most appropriate model.

Mobile Application:

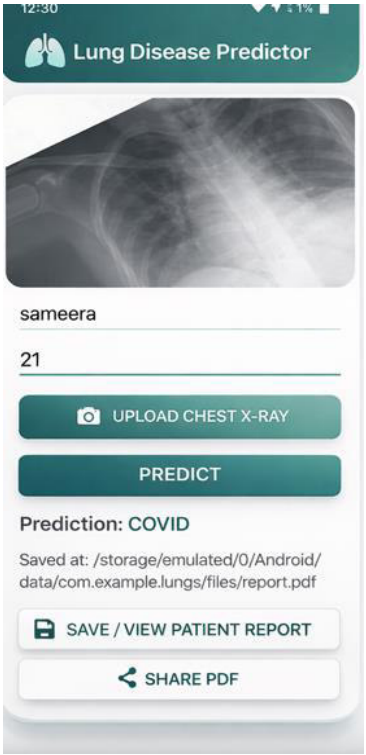


Figure 4.17 *User Interface of Lung Disease Predictor for X-ray Image Upload*

Figure 4.4 below highlights the user interface used in uploading the images as well as diagnosing any disease from the X-ray pictures. The user interface is simple, easy to understand and use by both medical practitioners and the general users. User can easily fill all the required information about the patients including their names and ages, thus making sure that no important data is left out while analyzing the picture.

After uploading of the picture, the user interface analyzes the information and gives the prediction results of the disease detected from the X-ray images. Apart from giving the prediction results, the user interface allows the option of saving the report generated, viewing the report and even sharing the report to health professionals.

In brief, the user interface ensures that the whole process of analyzing an X-ray picture and giving out prediction results is carried out easily.

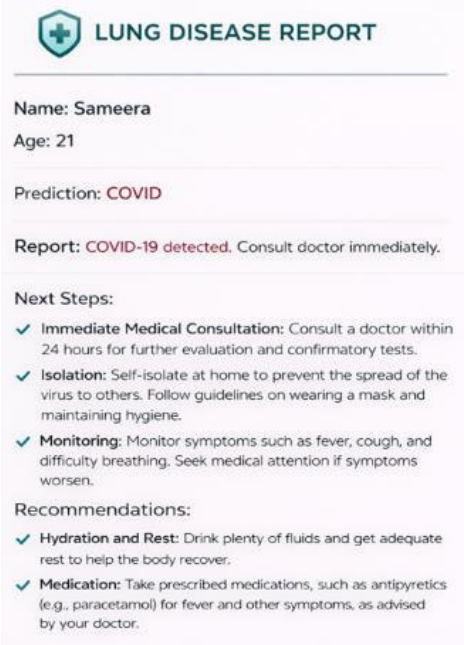


Figure 4.18: *Medical report for Patient*

The medical report produced using AI, as depicted in the figure below, features the patient's information, output predicted, and Grad-CAM visualization outputs. The medical report is extremely detailed, as it contains details of the disease identified and the level of confidence in the prediction. Moreover, the report gives recommendations on how the patient should be managed regarding his health status. Besides, the report suggests other possible measures that should be taken, such as more medical tests and consulting specialists for proper diagnosis. It also identifies critical measures that should be taken concerning the disease that has been identified.

Chapter-5

Conclusion and future Scope

5.1 Conclusion

The main aim of this project was to create a deep learning system which is capable of detecting and classifying lung diseases effectively through chest X-ray images. It was carried out in three phases - Mini-1, Mini-2, and the Major Project that each phase brought advancements in the accuracy, efficiency, and usability of the model.

At Mini-1 stage, the DenseNet201 deep learning method was used to build the main classification model. With this model, feature extraction and disease identification from chest X-ray images stages were done successfully. The outcomes pointed out that DenseNet201 was able to detect critical elements in medical images and give trustworthy classification results. But resource-wise this approach was pretty demanding so tweaking was in order to make the system more efficient.

At Mini-2 stage, with the help of EfficientNetB0 and Grad-CAM, the system was made better. By model re-scaling, EfficientNetB0 brought an enhancement to the classification effectiveness of the system while Grad-CAM gave us a visual understanding of the diseased part in the chest X-ray image through a heatmap. They not only got the results more transparent but also made the model more interpretable. In comparison with Mini-1, Mini-2 exhibited a significant advancement in the aspects of both prediction and visualization.

The cornerstone of the Major Project was implementing MobileNetV2 which served as the main deep learning architecture. The main reasons behind choosing this model were its compact design and excellent performance. Moreover, the model was incorporated into a mobile system that enabled instant illness diagnosis and the creation of reports. By drastically lowering the computational workload and increasing the prediction rate, MobileNetV2 made it feasible for the system to be run on a mobile device and used in real life. Through the GUI, the system was able to produce classification outputs, heatmaps, and written medical reports without any human intervention.

In general, the outcome kept getting better as the project continued. The comparison also revealed that the Major Project managed to surpass other works in terms of accuracy, speed, and user-friendliness. Combining deep learning methods with explainable AI tools not only improved the trustworthiness but also the transparency of the predictions.

As such, the entire model that has been designed turns out to be a strong and reliable mechanism for the automated detection of any kind of lung ailments using chest X-ray images. The model can be quite helpful to medical practitioners in diagnosing any kind of diseases at their early stages.

5.2 Future Scope

1. Actually, the designed system will strongly raise the classification accuracy and efficiency levels, but future improvements of the system are not excluded.

2. The major future improvements one can think of relate to significantly increasing the size and diversity of the dataset. By adding a much larger number of chest X-ray images from various hospitals and different patient groups, the model will be trained more effectively and accomplish a higher accuracy level. Besides that, it will significantly enhance the system's ability to generalize and minimize misclassification errors.

3. Another significant improvement for the future would be the introduction of more deep learning models like very advanced convolutional neural networks and transformer-based architectures, as these models bring better feature extraction and can even detect greater complexity in disease patterns.

4. Furthermore, the system can be designed to recognize different types of diseases apart from diseases detected in our project. It will prove to be a great asset as multi-class disease detection will greatly enhance the system's value in clinical settings and open more diagnostic avenues.

5. One other major benefit that will come with deployment of the system on cloud platforms is remote access from any location along with real-time sharing of diagnostic results among different hospitals and healthcare professionals.

6. Besides, the system can be connected to electronic health records (EHRs) which along with keeping patient history, will also be able to generate medical reports comprehensively and indirectly help doctors in monitoring patients over longer periods.

7. Adding more sophisticated explainable AI features to the system is one more way how it will gain in openness and it's a system people can rely on more. Besides giving medical

professionals tools for visualizing the model, there will be modules that provide comprehensive explanations to the users.

8. Besides that, this system could be developed to work with other types of medical images like CT scans and MRI pictures which means it will cover various diseases where diagnosis is done through images.

9. All things considered, this proposed system not only has tremendous potential to be developed further but also for implementation in the actual world. The system, armed with consistent advances in deep learning and medical facilities, may be turned into a very effective tool providing automated medical diagnostics and assisting with decision making.

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Appendix

Some screenshots of the code snippet:

```python
def extract_features(image):
    gray = cv2.cvtColor((image*255).astype('uint8'), cv2.COLOR_BGR2GRAY)

    # Normalize image to float32
    gray = gray.astype(np.float32) / 255.0

    # HOG Features
    hog_feat = hog(gray, pixels_per_cell=(16,16), cells_per_block=(2,2))

    # Sobel gradients (SAFE)
    sobelx = ndimage.sobel(gray, axis=0)
    sobely = ndimage.sobel(gray, axis=1)

    grad_mag = np.sqrt(sobelx**2 + sobely**2)

    #  Clip to avoid overflow
    grad_mag = np.clip(grad_mag, 0, 10)

    grad_var = np.var(grad_mag)

    # Ratios
    black_ratio = np.sum(gray < 0.2) / gray.size
    white_ratio = np.sum(gray > 0.8) / gray.size

    features = np.hstack([hog_feat, grad_var, black_ratio, white_ratio])

    #  FINAL SAFETY CHECK
    features = np.nan_to_num(features, nan=0.0, posinf=1.0, neginf=0.0)

    return features
```

Figure 1: Image Preprocessing and Feature Extraction

```python
def train_kmeans_sample(data_dir, classes, sample_size=500):

    paths = []
    for cls in classes:
        cls_path = os.path.join(data_dir, cls)
        for file in os.listdir(cls_path):
            paths.append(os.path.join(cls_path, file))

    np.random.shuffle(paths)
    paths = paths[:sample_size]

    features = []

    for path in paths:
        img = preprocess_image(path)
        feat = extract_features(img)
        features.append(feat)

    features = np.array(features)

    #  SCALE FEATURES (VERY IMPORTANT)
    scaler = StandardScaler()
    features = scaler.fit_transform(features)

    kmeans = KMeans(n_clusters=2, random_state=42)
    kmeans.fit(features)

    return kmeans, scaler
```

```python
kmeans = train_kmeans_sample("/content/drive/MyDrive/Lung Disease Dataset/train", classes, 500)
```

Figure 2: K-Means Training Function

```python
    def __getitem__(self, idx):
        batch_paths = self.paths[idx*self.batch_size:(idx+1)*self.batch_size]
        batch_labels = self.labels[idx*self.batch_size:(idx+1)*self.batch_size]

        images = []

        for path in batch_paths:
            img = preprocess_image(path)

            feat = extract_features(img).reshape(1, -1)
            feat = self.scaler.transform(feat)   #  IMPORTANT

            cluster = self.kmeans.predict(feat)[0]

            if cluster == 1:
                img = enhance(img)

            images.append(img)

        return np.array(images), tf.keras.utils.to_categorical(batch_labels, 5)
```

```python
train_dir = "/content/drive/MyDrive/Lung Disease Dataset/train"
val_dir   = "/content/drive/MyDrive/Lung Disease Dataset/val"
test_dir  = "/content/drive/MyDrive/Lung Disease Dataset/test"
```

Figure 3: Data Loading and Preprocessing in Batch Generation

```python
from tensorflow.keras.layers import GlobalAveragePooling2D, GlobalMaxPooling2D
from tensorflow.keras.layers import Dense, Reshape, multiply, Conv2D, Concatenate, Lambda

def cbam_block(input_feature, ratio=8):

    channel = input_feature.shape[-1]

    # ===== Channel Attention =====
    avg_pool = GlobalAveragePooling2D()(input_feature)
    max_pool = GlobalMaxPooling2D()(input_feature)

    shared = Dense(channel // ratio, activation='relu')

    avg_out = shared(avg_pool)
    max_out = shared(max_pool)

    channel_attention = Dense(channel, activation='sigmoid')(avg_out + max_out)
    channel_attention = Reshape((1,1,channel))(channel_attention)

    x = multiply([input_feature, channel_attention])
```

Figure 3: CBAM Attention Module (Channel Attention)

```python
input_img = Input(shape=(224,224,3))

base_model = MobileNetV2(weights='imagenet', include_top=False, input_tensor=input_img)

for layer in base_model.layers:
    layer.trainable = False

x = base_model.output
x = cbam_block(x)
x = GlobalAveragePooling2D()(x)

x = Dense(512, activation='relu')(x)
x = Dropout(0.5)(x)

output = Dense(5, activation='softmax')(x)

model = Model(inputs=input_img, outputs=output)

model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)
```

Figure 4: Code Snippet of MobileNetV2 Model

```python
def show_fig8(generator, n=3):
    batch_x, _ = generator[0]

    plt.figure(figsize=(12, 4*n))

    for i in range(n):
        real = batch_x[i]
        gen = enhance(real)
        rec = reconstruct(gen)

        plt.subplot(n,3,3*i+1)
        plt.imshow(real)
        plt.title("Real")
        plt.axis('off')

        plt.subplot(n,3,3*i+2)
        plt.imshow(gen)
        plt.title("Generated")
        plt.axis('off')

        plt.subplot(n,3,3*i+3)
        plt.imshow(rec)
        plt.title("Reconstructed")
        plt.axis('off')

    plt.tight_layout()
    plt.show()

show_fig8(test_gen)
```

Figure 5: Code Snippet of MobileNetV2 Architecture

```python
import numpy as np
from sklearn.metrics import classification_report

# Get true labels from the test generator
y_true = []
for i in range(len(test_gen)):
    _, labels = test_gen[i]
    y_true.extend(np.argmax(labels, axis=1))

# Get predictions from the model on the test generator
y_pred_raw = model.predict(test_gen)
y_pred = np.argmax(y_pred_raw, axis=1)

# Generate and print the classification report
report = classification_report(y_true, y_pred, target_names=classes)
print(report)
```

Figure 6: Code Snippet for Model's Classification Report

```python
import matplotlib.pyplot as plt

# Your values from training logs
train_acc = [
    0.6100, 0.7718, 0.7893, 0.8156, 0.8298,
    0.8385, 0.8519, 0.8437, 0.8684, 0.8437,
    0.8446, 0.8661, 0.8654, 0.8635, 0.8836,
    0.8853, 0.8915, 0.8919, 0.9079, 0.8962
]

val_acc = [
    0.7101, 0.7773, 0.7911, 0.8207, 0.8296,
    0.7931, 0.7857, 0.8440, 0.8588, 0.8528,
    0.8212, 0.8548, 0.8519, 0.8528, 0.8440,
    0.8548, 0.8519, 0.8583, 0.8622, 0.8598
]

epochs = range(1, 21)

plt.figure(figsize=(8,5))
plt.plot(epochs, train_acc, marker='o', label='Train Accuracy')
plt.plot(epochs, val_acc, marker='o', label='Validation Accuracy')

plt.title('Model Accuracy (Your Training)')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()
plt.grid()

plt.show()
```

Figure 7: Code Snippet for Training and Validation Accuracy

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